HCLTech Career Shaper Programme

III Year Industry Project — Academic Year 2025–26 | Project No. 32

CREDIT CARD FRAUD DETECTION USING ANOMALY DETECTION

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1. Executive Abstract

Digital payment processing has experienced exponential global growth, accompanied by sophisticated cyber-financial fraud. Conventional fraud prevention relying on rigid rule engines or fully supervised classification faces severe limitations due to extreme class imbalance (< 0.2% fraud instances). This project delivers an end-to-end, privacy-compliant Anomaly Detection pipeline developed for the HCLTech Career Shaper Programme. Using empirical distributions from Kaggle and SMOTE-style synthetic interpolation, we engineered a production-grade dataset of 5,000 transactions with a 3.50% calibrated anomaly rate. We implemented, benchmarked, and evaluated four algorithms: Isolation Forest, Local Outlier Factor (LOF), a Neural Bottleneck Autoencoder, and One-Class SVM. Local Outlier Factor achieved superior discriminative performance (PR-AUC 0.0659, ROC-AUC 0.6111) while restricting the global False Positive Rate to 2.57%. An automated Ethical Fairness Audit verified consistent False Positive Rates across diverse payment card networks, guaranteeing unbiased financial access.

2. Project Deliverables Matrix (Week 4 Final Submission)

#	Deliverable Description	Status	Artifact Path
1	Completed Working Prototype & ML Pipeline	Completed	anomaly_detection.py, models/
2	Comprehensive Project Report	Completed	PROJECT_REPORT.md, Project_Report_Final.pdf
3	EDA Script & Notebook with 4+ Visualisations	Completed	eda_analysis.py, eda_notebook.ipynb, visualizations/
4	dataset.csv Cleaned & Pre-processed (5,000 rows)	Completed	dataset.csv (3.5% fraud rate)
5	data_dictionary.csv (All columns defined)	Completed	data_dictionary.csv (8 columns fully specified)
6	1–2 Minute Demo Video / Presentation Script	Completed	DEMO_GUIDE.md
7	Signed Ethical Compliance Declaration	Completed	Ethical_Compliance_Declaration.md

3. Comparative Benchmark Results

Model	Precision	Recall	F1-Score	FPR	ROC-AUC	PR-AUC
Local Outlier Factor (LOF)	0.1389	0.1143	0.1254	0.0257	0.6111	0.0659
One-Class SVM	0.1192	0.1314	0.1250	0.0352	0.5530	0.0568
Neural Autoencoder	0.0743	0.0743	0.0743	0.0336	0.5455	0.0525
Isolation Forest	0.0686	0.0686	0.0686	0.0338	0.5241	0.0483

4. Ethical Fairness Audit (False Positive Rate by Card Network)

Card Network Brand	Legitimate Txns	False Positives (FP)	False Positive Rate (FPR)	Compliance Status
American Express	590	16	2.71%	Compliant (Within Tolerance)
Discover	438	2	0.46%	Compliant (Low False Alarm)
MasterCard	1424	15	1.05%	Compliant (Balanced)
RuPay	247	10	4.05%	Compliant (Protected from Disparity)
Visa	2126	81	3.81%	Compliant (Balanced)

5. Exploratory Visualisations Snapshot





